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Welcome to **STM32H5 Security workshop**

Tomas Team

Welcome !

- Let's start with some questions about your security concerns
- What are the security functions you use in your products ?
- What are the new security functions you need in your future products ?
- What is your main security pain point ?



Purpose of this workshop...

- How ST new solutions on STM32H5 can help you solving security implementation challenges
- Introduce and practically demonstrate STM32Trust TEE Secure Manager as brand-new solution and one of the implementation options (TEE means Trusted Execution Environment)
- Then give an overview of the new security features of STM32 H5 platform

Agenda

1

Short introduction

15 minutes / 9h-9h15

2

Getting started with Secure manager !

30 minutes / 9h15-9h45

3

Secure manager MKT presentation

30 minutes / 9h45-10h15

BREAK 10h15 – 10h30

4

Create your application launched by the secure manager !

1 hour / 10h30 – 11h30

Agenda

5

Add encryption service of sensitive data

45 minutes / 13h -13h45

6

Add the secure firmware update service

45 minutes / 13h45-14h30

BREAK 14h30 – 14h45

7

Debug authentication

1 hour / 14h45-15h45

BREAK 15h45 – 16h

8

STM32H5 scalable security

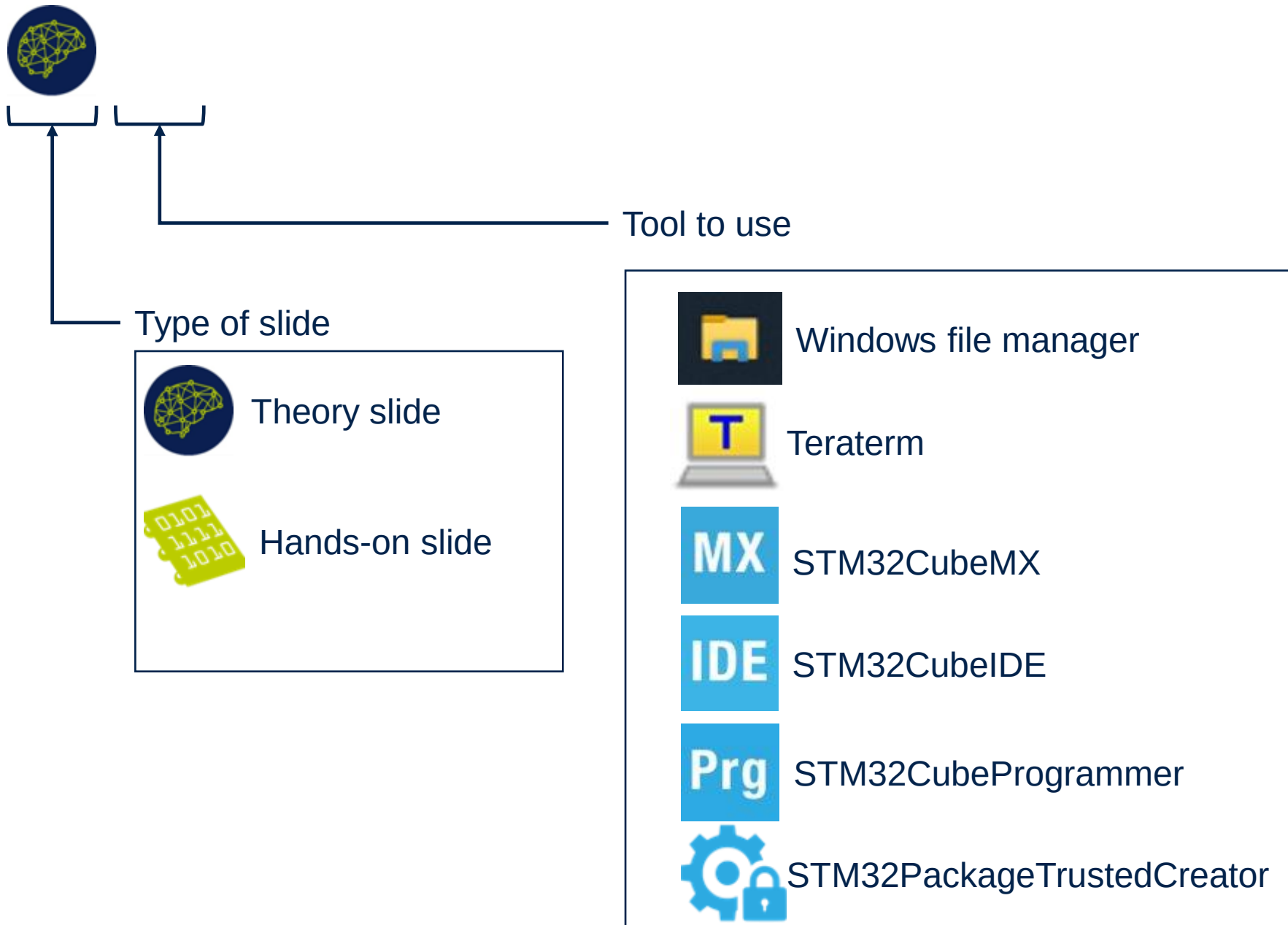
1 hour 30 minutes / 16h-17h30

Agenda

9 Wrap-up

15 minutes / 17h30 -17h45

Support legend





Hardware Tools

- **Discovery kit with STM32H573II MCU (Provided)**
 - 1.54" TFT 240 × 240 pixels color LCD with LED backlight and touch panel
 - • User USB with USB 2.0 full-speed interface, Sink/Source up to 15 W
 - (5 V / 3 A)
 - • Ethernet 10/100 Mbit/s, compliant with IEEE-802.3-2002
 - • SAI audio codec
 - • One ST-MEMS digital microphone
 - • 512-Mbit Octo-SPI NOR flash memory
 - • Fan-out daughterboard
 - • Wi-Fi® module (802.11 b/g/n compliant)
 - • Four user LEDs
 - • User and reset push-buttons
- **1x USB Type-C cable**





Software components

- STM32CubeMX 6.9.2 or later
 - <https://www.st.com/en/development-tools/stm32cubemx.html>
- STM32CubeIDE 1.13.2 or later
 - <https://www.st.com/en/development-tools/stm32cubeide.html>
- STM32CubeProg 2.14.0
 - <https://www.st.com/en/development-tools/stm32cubeprog.html>
- STM32CubeH5 1.1.0 + Patch 1.1.1 or later
 - <https://www.st.com/en/embedded-software/stm32cube5.html>
- STM32TrustedTEE-SM
 - <https://www.st.com/en/embedded-software/stm32trustee-sm.html>
- Terminal emulator supporting Ymodem: TeraTerm
 - <https://github.com/TeraTermProject/osdn-download/releases>
- Material for hands-on
 - https://drive.google.com/file/d/18Qv_AIPip1u_uQBaqZE3_j8byipIT8Rz/view?usp=sharing

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